

Lab 4 – Security Monitoring with Security Onion

➤ Learning Objectives

By the end of this lab, students should be able to:

- Understand how security tools are integrated into a SOC
 - Analyze alerts generated by **Suricata**
 - Investigate network activity using **Zeek** logs
 - Explore data through dashboards powered by **Elastic Stack**
 - Correlate multiple data sources
 - Reconstruct a security incident
-

➤ Prerequisites

Students must have:

- Completed Suricata and Zeek labs
 - Access to a Security Onion instance (VM recommended)
 - A PCAP file
-

➤ Lab Scenario

You are part of a **Security Operations Center (SOC)**.

A suspicious network capture has been provided.

Your mission:

- Analyze the traffic
- Detect suspicious activity
- Investigate and explain the incident

Part 1 : Platform Exploration

Mission 1: Discover the SOC Interface

Tasks

- Access Security Onion dashboards
 - Identify:
 - Alerts interface
 - Log search tools
 - Visualization dashboards
-

Part 2 : Alert Analysis (Suricata)

Mission 2: Identify Alerts

Tasks

- Locate alerts generated by **Suricata**
 - Identify:
 - Alert types
 - Frequency
 - Select one alert and extract:
 - Source IP
 - Destination IP
 - Protocol
 - Alert message
-

Part 3 : Deep Investigation (Zeek Logs)

Mission 3: Correlate with Network Logs

Tasks

Using **Zeek logs**:

- Investigate:
 - connections
 - DNS queries
 - HTTP activity
- Find activity related to the alert

Part 4 : Build an Incident Story

Mission 4: Reconstruct the Incident

Tasks

Using all collected data:

- Describe:
 - what happened
 - who initiated the activity
 - how the communication occurred
- Build a timeline of events

Deliverable

- Incident report (10–15 lines)
-

Tips & Best Practices

- Do not trust alerts blindly; **investigate them!**
 - Always correlate: alerts - logs
 - Focus on:
 - patterns
 - timelines
 - relationships between events
 - Think like: *“What is the attacker trying to do?”*
-

Reflection Questions

- What is the role of a SOC analyst?
- Why is correlation between tools essential?
- What is the difference between detection and investigation?
- Why are alerts not enough on their own?
- What are the limitations of a monitoring platform like Security Onion?

“Tools don’t stop attacks. Analysts do.”